Install and set up SI Water

SI Water runs your borehole pump and municipal backup valve from one pressure probe in the tank. All the decisions are made on the controller, so it keeps working when the internet is down. This guide takes you from mounting to a tank that fills itself.

In the box

- SI Water controller in an IP65 enclosure, with circuit breaker and 12 V DIN-rail supply
- Stainless pressure probe on a 5 m cable, already wired to the controller (up to 20 m on request)
- Cable glands for the mains, the probe and the pump signal
- This guide

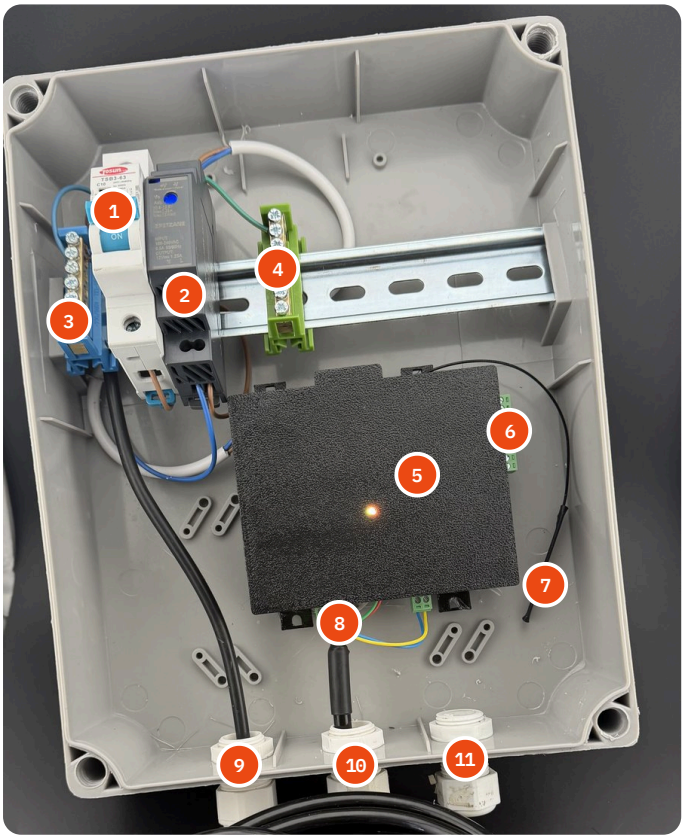
You'll need

- A qualified electrician to bring in 230 V and connect the pump signal
- 2.4 GHz WiFi that reaches the enclosure
- The Solaire Intelligence app (iPhone or Android) and an account
- The tank's height and volume

1 Mount the enclosure

- Mount on a wall near the tank, out of direct rain and spray, at a comfortable working height.
- Cables enter through the glands under the box: mains in, the probe cable, and the pump signal. Fit one more gland for the valve if the site has municipal backup.
- The probe cable is 5 m, so the enclosure goes within reach of the tank.

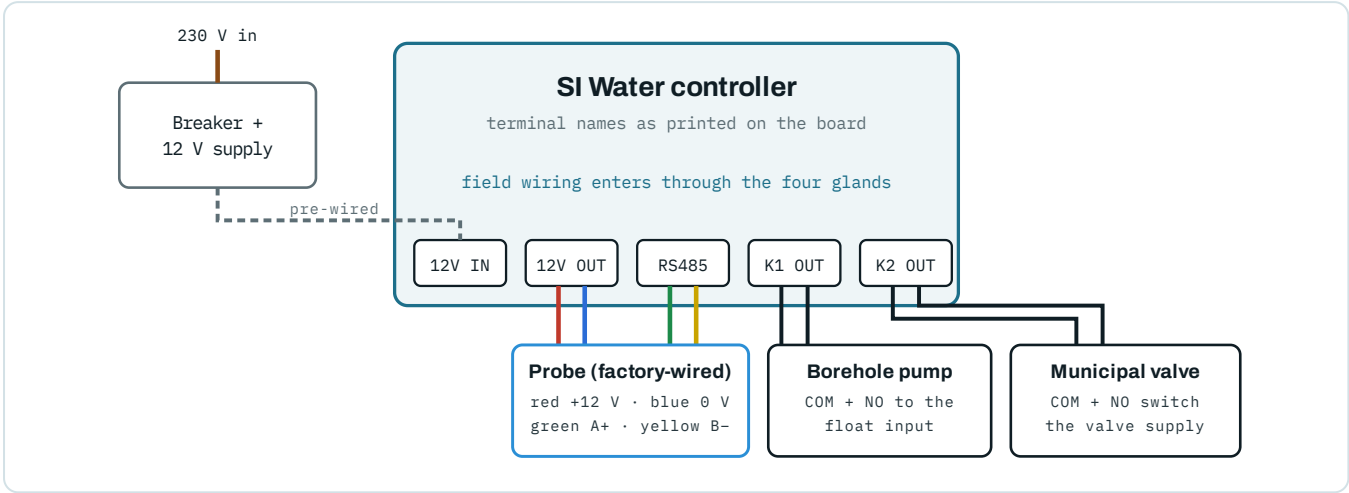
2 What's inside



1. **Circuit breaker.** Switches the whole unit.
2. **12 V supply.** Powers the controller and the probe.
3. **Neutral bar**
4. **Earth bar**
5. **SI Water controller.** The status light shows through the case (section 9).
6. **K1 and K2 outputs.** The pump signal and the municipal valve connect here.
7. **WiFi antenna.** Leave it free, away from metal.
8. **Probe wires,** factory-connected.
9. **Mains entry** for the electrician's 230 V cable.
10. **Probe cable** gland.
11. **Pump signal** gland. Add one for the valve if needed.

3 Connect the pump and valve

Isolate the supply first. The breaker, power supply, controller and probe are wired at the factory. On site the electrician connects the 230 V supply, the pump signal on K1 and, if used, the municipal valve on K2.



TERMINAL	CONNECT	NOTES
Breaker	230 V supply (L, N, E)	Electrician's work. Isolate before wiring.
12V OUT	Probe red (+12 V), blue (0 V)	Factory-wired. Only touch if the probe is replaced.
RS485	Probe green (A+), yellow (B−)	Factory-wired. If a replaced probe never reads, swap A and B.
K1 OUT	Borehole pump controller	Volt-free COM and NO. Use the input the float switch used. Never the motor directly.
K2 OUT	Municipal fill valve	Volt-free COM and NO. Leave empty if there is no municipal supply.

4 Install the probe

- The probe comes wired in. Uncoil the cable and lower the probe until it rests on, or just above, the tank floor. It measures the water above its tip.
- Tie the cable off at the tank lip so the probe can't drift or be pulled up.
- The cable carries a thin air tube. Its end must stay dry inside the enclosure: don't seal, kink or submerge it.
- Keep the probe cable away from mains cables where you can.

5 Switch on and connect

1. Switch on the breaker. The status light comes on amber while the controller starts.
2. In the Solaire Intelligence app, tap **Add device**.
3. When the app asks, join the WiFi network `SI-Water-Setup` with the password `solairesetup`.
4. Pick the property's WiFi from the list the controller shows, and enter its password. It must be 2.4 GHz.
5. Choose the site the unit belongs to. Installers pick the client first.
6. The controller joins the WiFi, registers itself and restarts. The light turns green when it has registered, and SI Water appears on the dashboard.

Phone hotspot as the WiFi? On an iPhone, turn on **Maximise Compatibility** under Personal Hotspot so it runs on 2.4 GHz. The phone running the hotspot can't also join SI-Water-Setup, so do the setup from a second phone or a laptop.

Setting up without the app. Join SI-Water-Setup, browse to `192.168.4.1`, fill in **Setup: WiFi SSID**, **Setup: WiFi Password** and **Setup: Client ID** (the 6-character site code in the app), then press **Setup: Claim Device**.

Red light after setup? The registration failed, usually a mistyped WiFi password. The setup network stays up, so run the steps again.

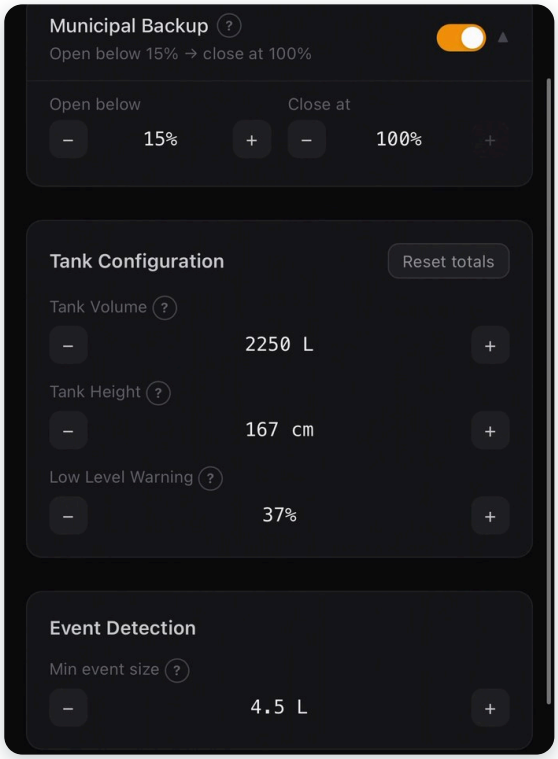
6 Tell it about the tank

In the app, open SI Water, go to **Settings** and find **Tank Configuration**. Set these first, so every litre and percentage reads true.

SETTING	SET TO	DEFAULT
Tank Height	Water depth when full: probe tip up to the overflow level, in cm	167 cm
Tank Volume	Usable litres at that height. The rated size is fine	2250 L
Low Level Warning	Level that raises the Low Tank warning	20%
Min event size	Smallest draw counted as a water event	6 L

Check it once. Compare Tank Level with a dipstick or sight glass at one known level. Two minutes now saves every reading after. Tanks that aren't straight-sided are set up per project; ask us.

Probe Fault means the controller hasn't had a reading for a minute. Check the probe's power and the A/B wires. In Auto Mode, filling stops and both outputs switch off until the reading returns.

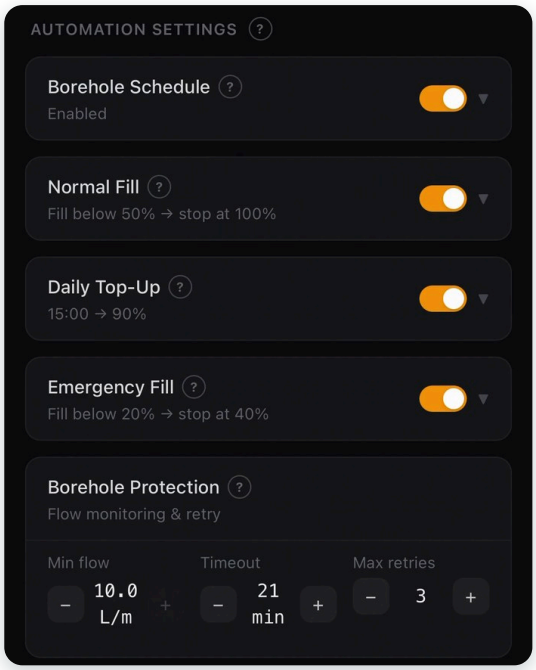


Settings: municipal backup, tank configuration and event detection.

7 Set up filling

A new unit switches nothing until **Auto Mode** is on. Turn on the modes this site needs, check their levels, then switch Auto Mode on last.

MODE	WHAT IT DOES	DEFAULT
Normal Fill	Runs the borehole when the tank falls to the start level, stops at the stop level. The everyday mode.	off · 60% → 80%
Borehole Schedule	Limits normal, emergency and daily top-up filling to a time window. Critical Override ignores it.	off · 06:00–18:00
Daily Top-Up	Once a day at a set time, fills to the target if the tank is below it.	off · 07:00 → 80%
Emergency Fill	Fills when the level falls below its start level, inside the schedule window if one is set.	on · 30% → 80%
Critical Override	Below this level it fills immediately, ignoring the schedule.	15%
Municipal Backup	Opens the municipal valve when the borehole is off or has faulted.	off · 20% → 60%



Settings: each mode has its own switch and levels. Tap a mode to change it.

> **Order of priority:** Critical, then Emergency, then Normal Fill, then Municipal. In Auto Mode the borehole pump and the municipal valve never run together. At 99% both outputs switch off, in every mode.

8 Pump protection

While the borehole runs, the controller watches how fast the tank is rising. If the borehole stops delivering, the pump is switched off and rested before trying again.

SETTING	MEANING	DEFAULT
Minimum Flow Rate	Below this inflow, the pump counts as not delivering	0.5 L/min
No Flow Timeout	How long low flow is allowed before the pump is stopped	5 min
Maximum Retry Count	Restarts before it locks out. The rests between tries are 20, 30, then 60 minutes	3

> After the last retry, **Borehole Fault** latches and the pump stays off. Municipal Backup takes over if it's on. Once the borehole has recovered, press **Borehole Fault Reset** in the app. A borehole that trips repeatedly is telling you the water table has dropped: don't just raise the retry count.

9 The status light

LIGHT	MEANING
Short blue flash every 3 s	Running normally, nothing filling
Steady blue	Borehole pump running, or a firmware update in progress
Steady cyan	Municipal valve open
Magenta, flashing	Low flow from the borehole. Pump resting before a retry
Amber, flashing	Probe Fault: no level reading
Red, flashing	Borehole Fault. Check the borehole, then press Borehole Fault Reset
Steady amber	Starting up, or waiting to be set up

10 Warnings in the app

WARNING	MEANING AND WHAT TO DO
Low Tank Warning	Level is below your low-level setting. Filling should already be running.
Low Flow Warning	The borehole isn't delivering while it runs: water table, valve or pipe restriction.
Borehole Fault	Pump protection locked out after all retries. Investigate, then Borehole Fault Reset.
Critical Warning	Tank at critical level. Check the water supply now.
Probe Fault	No level reading for a minute. Check the probe's power and A/B wiring.

Water System Status shows what the controller is doing in words: Idle, Normal Fill, Daily Top-Up, Emergency Fill, Critical Fill, Retry Waiting, Municipal Fill, Manual, or a fault.

11 Looking after it

Firmware updates

When an update is ready, the Firmware card on the SI Water page shows an **Update** button. The controller stops the pump and valve, updates itself and carries on. No site visit.

New WiFi network

Use **Network Config** in the app. The controller opens its setup network again so you can choose the new WiFi. Its settings and history stay as they are.

Manual control

With Auto Mode off you can switch the borehole pump and municipal valve by hand from the app. The overflow guard still applies.

Factory reset

Hold the **BOOT** button on the controller for 10 seconds. The light turns bright magenta, the unit forgets its WiFi and its site, and restarts ready to be set up again.

Support. Solaire Intelligence · `sales@solairesa.co.za` · `073 894 8248`. Quote the serial number shown as **SI Serial** on the SI Water page.